# Mapping the Epstein Aviation Network: A Provenance Analysis

## Contents

# 1 Introduction

The release of the Epstein Files over the past two years has given the public access to over 3 million pages of documents, email, deposition transcripts, and media gathered over two decades of federal investigations into the late financier and convicted sex offender Jeffrey Epstein(Associated Press, 2024).

Public interest has focused heavily on the broader network associated with Epstein, particularly high profile figures from the worlds of entertainment and politics. Central to this narrative has been the story of Epstein's island, and people being flown in on his private jet. This dataset provides an opportunity to examine this social circle through the relationships recorded in the released documents, which will be referred to hereafter as the aviation network.

Constructing meaningful relationship networks from these sources presents several methodological challenges. Relationships within the corpus originate from very different contexts ranging from formal legal testimony to second-hand media reporting. In building the network, we will retain details of the source documents to account for this. The use of multiple aliases for the same individual e.g. President Clinton vs. Bill Clinton posed a second challenge. Identifying these aliases and consolidating entities is a key step. In addition multiple claims can originate from a single document which if counted by the claim can inflate connections. In casting a wide net for the aviation network we will also inevitably encounter noise. We will take two approaches to filter for noise- first removing entities which are not people, and secondly manually comparing the people returned with external sources to ensure that the people in our final network have not been mistakenly included from a different context. Finally, given the source material clearly Jeffrey Epstein is the core entity, however while we will use him to build the networks, we will not include him in the aviation network as we are interested in the structure of this social circle beyond their common connection to Epstein.

The report therefore aims to

- identify and curate the key entities within the aviation network based on the Epstein Files.
- analyse the structural characteristics of the resulting network.
- compare the full network with a legal-document subnetwork derived from legal-source material.

There is no suggestion that appearing in these documents implies wrongdoing. Many individuals who appear within the released material have denied any wrongdoing in relation to Epstein, and the networks presented here should be interpreted as representations of relationships extracted from the source corpus rather than evidence of criminal activity.

## 2 Building the Aviation Network

### 2.1 Source Data and Graph Structure

The original graph is stored in a memgraph database containing 546,648 nodes and 1,432,277 edges. The nodes of interest to us are entities (people), documents and claims and the associated entity to entity edge related to. The inclusion of document and claim identifiers at the edge level allows relationships to be traced back to their supporting source material throughout the analysis. Limiting the analysis to these node and edge types reduces the graph to 187,167 nodes and 106,812 edges.

**Table 1:** Node and edge types used in the analysis

| Element | Key attributes |
|---|---|
| Entity | name |
| Document | doc_id, category, paragraph_summary |
| Claim | claim_id |
| RELATED_TO | action, doc_id, claim_id |

### 2.2 Identifying Aviation Entities

The first stage of the analysis was to identify entities connected to Jeffrey Epstein through aviation-related relationships. These entities formed the initial pool of candidates for the aviation network. A predefined set of aviation-related actions was used to filter relationships involving Epstein. A sample of the action categories and examples are provided in Table 2.

**Table 2:** Categories of aviation-related actions used to identify candidate actors

| Search Category | Example Actions |
|---|---|
| Travel | traveled with, traveled internationally with, traveled on private jet with, traveled by private jet with |
| Flights | flew with, flew on private jet of, flew internationally with, flew as passenger on |
| Transportation | transported via private jet, transported with, transported to, transported victims via |
| Aircraft Provision | provided private jet travel to, provided aircraft to, offered jet transport to, chartered private aircraft for |
| Aircraft Participation | boarded aircraft with, arrived at private aircraft with, invited to fly on private jet |
| Flight Logs | appears on flight logs with, appeared in flight logs of, appeared on flight log with |
| Recruitment and Transport | recruited and transported to, recruited and transported minors for, coordinated recruitment and transport to |

The resulting extraction returned 290 edges between 92 entities. The next stage was removing entities which were not people. Examples are given in Table 3.

**Table 3:** Categories of terms removed during noise filtering

| Category | Examples |
|---|---|
| Aircraft and Transport Objects | jet, plane, aircraft, helicopter |
| Locations and Properties | island, ranch, apartment, property, real estate, London, multiple properties |
| Occupations and Roles | driver, therapist |
| Events and Activities | conference, dinner event |
| Organisations and Businesses | modeling agency |
| Legal or Contextual References | victims' rights, sexual assaults |

This reduced the graph to 264 edges between 69 people. The final step of this section was to consolidate people by their aliases. For example 'Virginia L. Giuffre','Virginia Roberts Giuffre','Virginia'). Entities were all mapped to their canonical name; however, the aliases were retained for future searches. This resulted in 264 edges between 60 people.

## 2.3 Manually pruning the Actor Set

Reducing the graph from 546,648 nodes to just 60 candidate aviation entities made a manual review feasible. Inspection of these entities revealed a mixture of genuine actors, anonymous placeholders ('Unknown Person B'), composite entities ('Ghislaine Maxwell, Virginia Roberts, and ET'), and extraction artefacts('ay') that required further curation before network construction could proceed. Manual pruning followed three main criteria. First, the entity had to correspond to an identifiable individual rather than a placeholder, generic reference, location, object, or text fragment. For this reason, entries such as Unknown Person B', Witness', Defendant', and ay' were removed. Second, the entity had to represent a single person rather than a composite extraction. Composite entries such as 'Ghislaine Maxwell, Virginia Roberts, and ET' or 'Bill Clinton, Kevin Spacey, Chris Tucker' were removed because they could not be treated as stable person-level nodes. Third, the entity had to be relevant to the aviation network being analysed here rather than a historically separate or weakly contextualised Epstein relationship. For example, Steven Hoffenberg was excluded because the aviation relationship referred to Epstein flying on Hoffenberg's jet in the late 1980's (Shanahan, 2022)- a different context from the social circle explored in this report. These criteria reduced the 60 candidate aviation entities to a curated set of 23 identifiable individuals.

## 2.4 Reconstructing Actor Relationships

The aviation network constructed in the previous sections identifies a curated set of actors associated with Jeffrey Epstein through aviation-related interactions. However, restricting analysis to aviation relationships alone would produce a network that largely reflects travel activity rather than the broader relationships captured within the document corpus. The next stage of the analysis therefore reconstructed all extracted relationships among the curated actor set (including all recorded aliases), regardless of relationship type. For each entity-pair of the curated actor set, all edges recorded within the graph database were retrieved. At this stage the aviation action filter was removed entirely, allowing political, social, legal, financial, and communication-related interactions to be recovered. This maintained our 23 nodes but expanded our edges from 264 to 2,761.

### 2.5 Removing Epstein

The aviation extraction process used Jeffrey Epstein as the anchor node for identifying candidate actors. Retaining Epstein in the final network would therefore guarantee that almost every actor remained connected through their relationship to a single highly connected individual. While this would produce a more connected graph, it would provide limited insight into the relationships among the actors themselves. Epstein was therefore removed prior to final network construction. Removing Epstein shifts the focus of the analysis from actor-Epstein relationships to actor-actor relationships. The resulting network therefore captures connections among members of the curated actor set rather than their individual association with Epstein. The consequences of this however are that certain individuals have no documented relationships within the aviation network except through Epstein. As a result, they are removed from the network, leaving us with a final network of 18 actors. They have been grouped into three categories: **Epstein Associates** contains people who either worked for Epstein directly or were involved in recruitment for his network. **Political Figures** is composed of individuals who have held public office in the US and internationally. **Celebrities** are well known people in the entertainment industry. A short description can be found in Table 4.

**Table 4:** Final aviation actor set used for network construction

| Name | Description |
|---|---|
| **Epstein Associates** | |
| Ghislaine Maxwell | Long-time associate and collaborator of Epstein |
| Jean-Luc Didier Henri-Rene Brunel | Modelling agent recruited for Epstein |
| Alan M. Dershowitz | Lawyer representing Epstein |
| Nadia Marcinkova | Assistant to Jeffrey Epstein |
| Adriana Mucinska | Assistant to Jeffrey Epstein |
| Sarah Kellen | Administrative assistant of Epstein |
| Virginia Roberts Giuffre | Prominent accuser and witness |
| **Political Figures** | |
| Al Gore | Former U.S. Vice President |
| Bill Clinton | Former U.S. President |
| Donald Trump | U.S. President |
| Ehud Barak | Former Prime Minister of Israel |
| Prince Andrew, Duke of York | Member of the British royal family |
| Sandy Berger | Former U.S. National Security Advisor |
| Lawrence Summers | Economist and former U.S. Treasury Secretary |
| **Celebrities** | |
| Chris Tucker | Actor and comedian |
| Heidi Klum | Model and television personality |
| Kevin Spacey | Actor and producer |
| Naomi Campbell | Model and media personality |

### 2.6 Edge Enrichment

To enrich and refine these connections, we used the doc_id, claim_id attributes of the edge to pull in document category for each edge. The original graph contained fourteen document categories. For analysis and visualisation purposes, these categories were consolidated into five broad provenance groups including an "other" category. The resulting grouping distinguishes between legal documents, media sources, direct communications, and financial records

**Table 5:** Document provenance groups used in the analysis

| Provenance Group | Original Categories |
| --- | --- |
| Legal | court_filing, legal_filing, transcript, public_record, public_records |
| Media | media_article, report, book_excerpt |
| Communication | email, letter, memorandum |
| Financial | financial_document |
| Other | mixed_document, other |

The resulting distribution across the four provenance groups can be seen in Figure 1. Media sources account for the largest proportion of supporting documents, outnumbering legal filings and direct communications, with financial documents a negligible contributor. Given the differing evidential standards associated with these source types, a parallel legal network was constructed to allow comparison between the full relationship network and one derived exclusively from legal-source material.

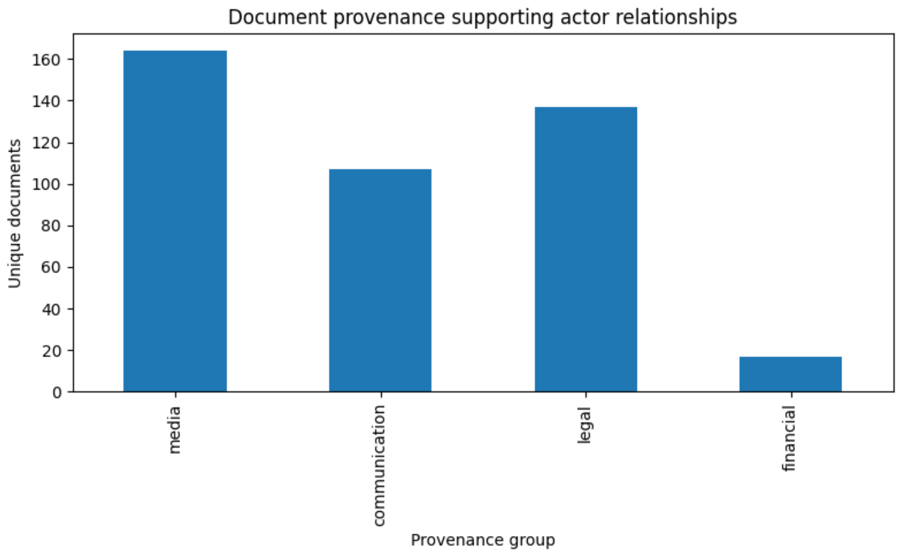


**Figure 1:** Distribution of Document Provenance in Aviation Network

## 2.7 Edge Aggregation

At this stage, relationships were still represented as individual events extracted from the source corpus. Multiple events could refer to the same pair of actors, often originating from the same document or repeated across multiple documents. To construct a network suitable for analysis, event-level relationships were aggregated into undirected actor pairs. Aggregation was performed at the document level rather than the claim level. This decision avoided inflating edge strength when multiple claims referring to the same relationship were extracted from a single source document. Each document therefore contributed at most one unit of support to a given actor pair. For each pair, the following edge attributes were associated:

- **Weight**: the number of distinct documents supporting a relationship.

- **Legal Weight**: the number of supporting documents belonging to the legal provenance group.
- **Legal Proportion**: the proportion of supporting documents belonging to the legal provenance group.

This resulted in 61 edges across the 23 nodes. Figure 2 shows us the distribution of **Legal Proportion** across all nodes. We can see that there is a significant number of relationships (25) which are documented in media/direct communication but have no legal record within this corpus, and that all levels of legal proportion are represented.

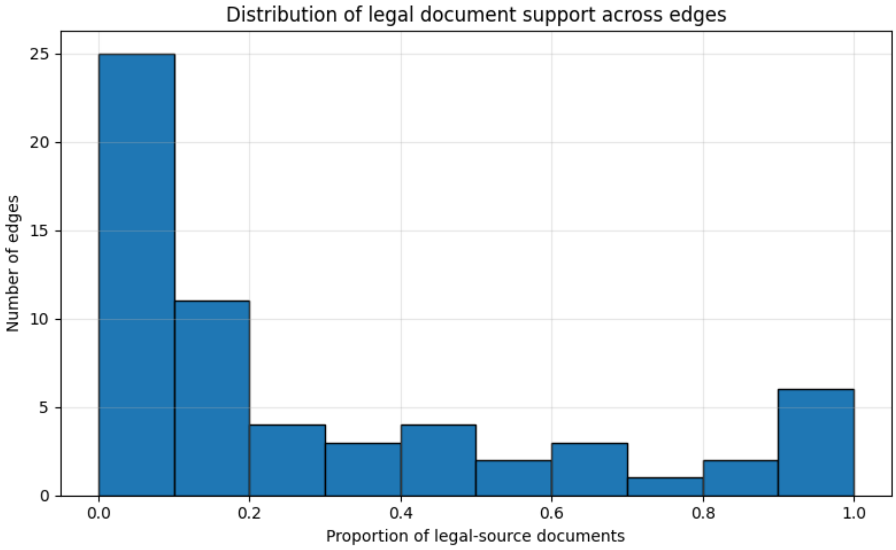


**Figure 2:** Distribution of Proportion of Legal Documents for each node-node relation

Before continuing to the analysis, it must be noted that the aggregation process relies on several assumptions. First, document counts are treated as a proxy for evidential support, although multiple documents may ultimately derive from the same underlying source. This is particularly relevant to media & email documents. Second, all documents within a provenance category are weighted equally, regardless of their length, detail, or reliability. Third, relationships are treated as undirected, meaning that distinctions between the initiator and recipient of an interaction are not retained. These assumptions should be considered when interpreting the resulting network measures.

# 3 Network Analysis

## 3.1 Overview of the Aviation Network

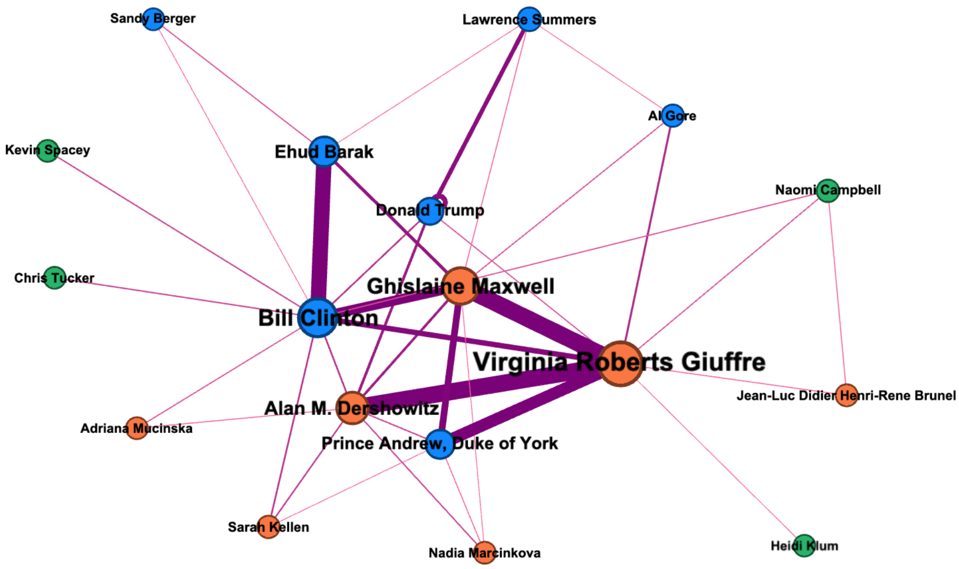


**Figure 3:** Aviation Network produced in Gephi

Figure 3 shows the reconstructed aviation network following the removal of Epstein and the aggregation of document-supported relationships. Edge thickness is determined by overall weight. Nodes are scaled by their PageRank and are coloured by category:

- Orange: Epstein Associates
- Blue: Political
- Green: Celebrity

Celebrities are (with the exception of Naomi Campbell) peripheral to the network, but there is a dense core with key political and Epstein associates intertwined. Table 6 summarises the structure of the final reconstructed network. The density of 0.248 means that roughly one quarter of all possible actor-to-actor connections are present in the network. For a social network, this indicates a relatively closely connected group rather than a sparse collection of isolated relationships. The average path length of 1.974 means that, on average, any actor can be reached from any other actor in approximately two steps. Together with the diameter of 3, this suggests that the network is compact: no actor is more than three relationship steps away from any other actor in the final graph (Wasserman & Faust, 1994). This suggests that the aviation-derived actor set is not simply a collection of independent Epstein-linked individuals. Once Epstein is removed, the remaining actors still form a fully connected structure with a dense central core. The network therefore captures relationships among the actors themselves, not only their shared relationship to Epstein.

**Table 6:** Summary statistics for the final reconstructed network

| Metric | Value |
|---|---|
| Nodes | 18 |
| Edges | 38 |
| Density | 0.248 |
| Diameter | 3 |
| Average path length | 1.974 |

### 3.2 Central Actors in the Full Network

**Table 7:** Top ten actors in the full network ranked by PageRank calculated using NetworkX

| Rank | Actor | Weighted degree | Betweenness | PageRank |
|---|---|---|---|---|
| 1 | Virginia Roberts Giuffre | 295 | 0.277 | 0.211 |
| 2 | Bill Clinton | 198 | 0.433 | 0.163 |
| 3 | Ghislaine Maxwell | 213 | 0.145 | 0.147 |
| 4 | Alan M. Dershowitz | 137 | 0.107 | 0.103 |
| 5 | Ehud Barak | 104 | 0.030 | 0.085 |
| 6 | Prince Andrew, Duke of York | 107 | 0.025 | 0.075 |
| 7 | Donald Trump | 80 | 0.026 | 0.057 |
| 8 | Lawrence Summers | 25 | 0.015 | 0.028 |
| 9 | Naomi Campbell | 8 | 0.021 | 0.018 |
| 10 | Sarah Kellen | 12 | 0.007 | 0.016 |

Virginia Roberts Giuffre, Bill Clinton, Ghislaine Maxwell and Alan M. Dershowitz emerge as the most prominent actors in the reconstructed network according to PageRank (Page et al., 1999). Bill Clinton exhibits the highest betweenness centrality, suggesting that he occupies an important bridging position between otherwise weakly connected parts of the network. Figure 3 also shows this, with Bill Clinton a bridge between celebrities, politicians and Epstein associates.

### 3.3 The Legal Network

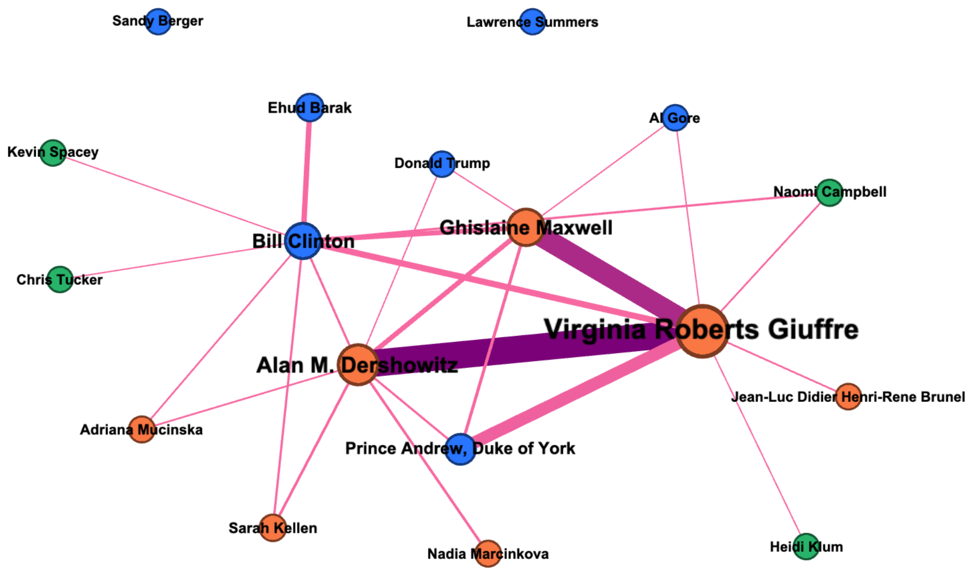


**Figure 4:** Legal Sub-network within Aviation Network

Reducing the connections exclusively to documents within the formal legal framework reduced the density slightly, from 0.248 to 0.208, with two political actors disconnected (no legal connections within the corpus). The remaining 16 actors remain fully connected with a comparable diameter and average path length to the original network. The legal subnetwork is smaller, but it is not fragmented. This suggests that legal-source documents preserve the core structure of the aviation network, even while removing a substantial number of weaker or non-legal relationships.

**Table 8:** Summary statistics for the legal network

| Metric | Value |
|---|---|
| Nodes | 16 |
| Edges | 25 |
| Density | 0.208 |
| Diameter | 3 |
| Average path length | 2.025 |

## 3.4 Comparing the Full and Legal Networks

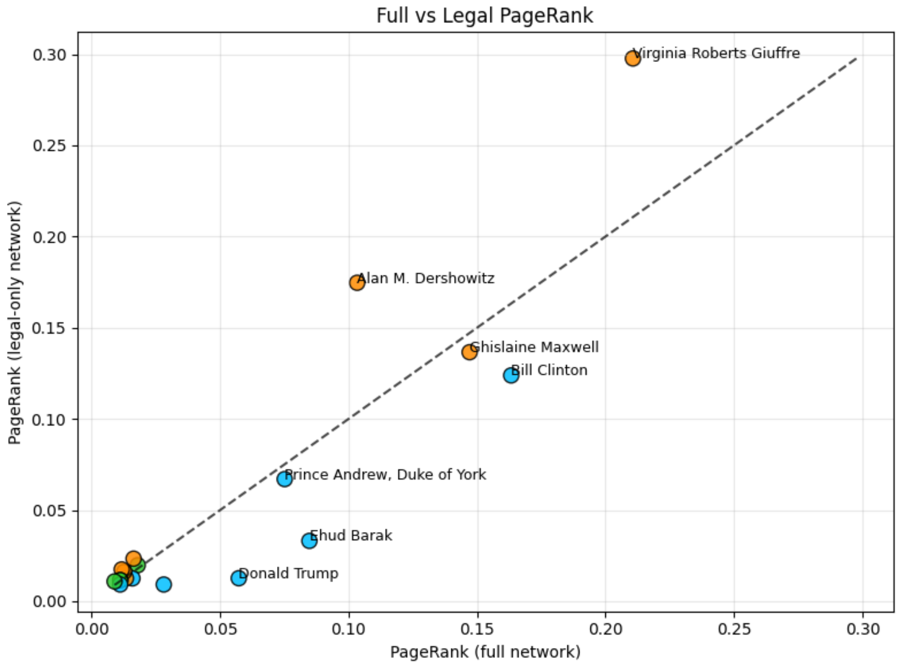


**Figure 5:** Comparing PageRank between full network and legal sub network

PageRank gives us an interesting perspective on the difference between the networks. In Figure 5 the dotted line $y = x$ represents a consistent PageRank, with entities above the line increasing in importance within the legal sub network and those below decreasing in importance. The two people with the highest legal PageRank also show the largest increase from the full network. They are Virginia Giuffre, the most prominent Epstein accuser and Alan Dershowitz, Epstein's lawyer. In contrast, a number of politicians decrease in relative importance. Donald Trump for example decreases significantly in page rank. Bill Clinton decreases by a similar amount in absolute terms but retains his overall importance within the network. This drop suggests that politicians' presence in the legal documentation is proportionally lower than that of their presence in the media and correspondence documentation.

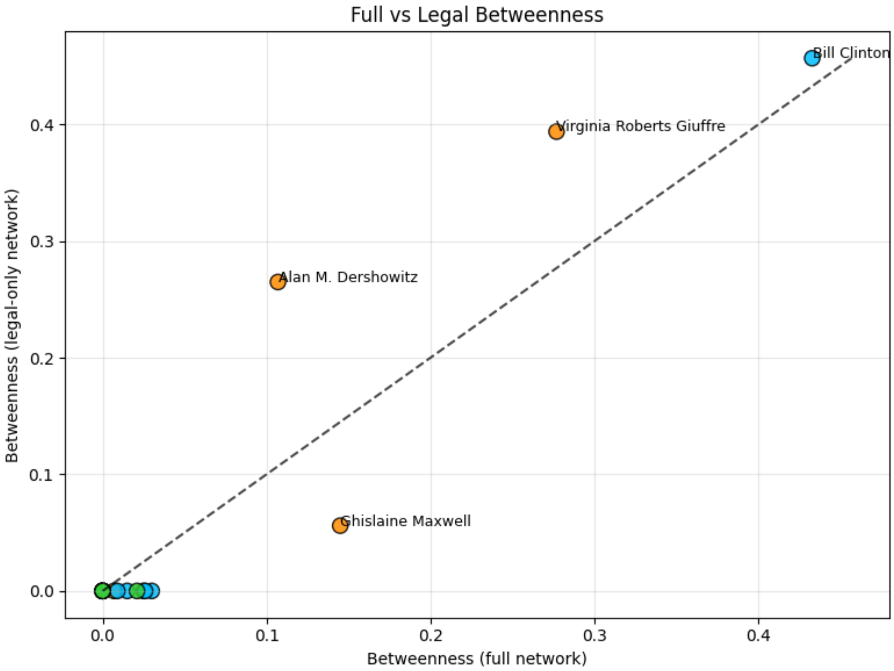


**Figure 6:** Comparing Betweeness between full network and legal sub network

Figure 6 shows most actors dropping to zero betweeness, indicating a more radial network focused on a few central actors. Those central actors remain the four dominant entites from the full network: Virginia Roberts Giuffre, Bill Clinton, Ghislaine Maxwell and Alan M. Dershowitz. In contrast to his drop in PageRank, Bill Clinton's centrality increases in the legal network, suggesting that although Clinton becomes less dominant in terms of direct legal-document support, he occupies a more important bridging position in the legal network. In other words, a larger share of the shortest paths between other actors pass through him once non-legal relationships are removed.

Structurally, this suggests that some of the media and communication edges in the full network provide alternative routes between actors. When those edges are stripped away, the legal network becomes more dependent on a smaller number of connecting nodes. Clinton's increased betweenness therefore does not mean that he is the most legally prominent actor overall, but that he helps connect otherwise more weakly linked parts of the legal subnetwork.

# 4 Limitations

There are several limitations to this analysis. The first is that this is a network built from extracted relationships in a document corpus, not a verified social network. An edge means that a relationship between two actors was identified in the source material and retained through the cleaning and aggregation process. It does not imply wrongdoing, direct collaboration, or even necessarily a strong real-world relationship.

A second limitation is the quality of the original extraction. The graph was built using an LLM to extract names and actions from a large and varied collection of documents. This made it possible to work at scale, but it also introduced noise. Some entities were extracted as fragments, composite names, anonymous placeholders, or aliases. Manual pruning and entity consolidation reduced this problem, but cannot remove it completely.

The edge weights should also be interpreted carefully. In this report, edge weight is based on the number of distinct documents supporting a relationship. This is useful because it avoids counting repeated claims from the same document multiple times. However, document count is only a proxy for support. Several documents may repeat the same underlying information, especially in media or email chains, while a single legal transcript may contain much richer information than several shorter documents.

The legal network should also not be treated as a ground-truth version of the network. A legal edge simply means that the relationship is supported by documents classified as legal, such as court filings, transcripts, or public records. Legal provenance gives a different evidential lens, but it does not automatically mean that the relationship is more accurate, more important, or legally established.

Finally, several modelling choices affect the final network structure. Relationships were treated as undirected, so the analysis does not distinguish between the initiator and recipient of an interaction. The actor set was also manually curated, which improved interpretability but introduced judgement into the construction process. The results should therefore be understood as a structured view of relationships present in the corpus, rather than a definitive account of the real-world Epstein network.

## 5 Future Work

There are several ways this analysis could be extended. The network could also be developed as a directed graph. In this report, relationships were treated as undirected because the focus was on whether two actors were connected within the corpus. However, direction may matter for certain relationship types. A directed model could preserve more of this structure and allow future analysis of roles within the network rather than only connections between actors.

A second extension would be to validate the aviation relationships more systematically against external sources. In this report, manual checks were used during actor curation, but a fuller analysis could cross-reference extracted relationships with official flight logs, court exhibits, or other structured records outside of the Epstein Files. This would help separate relationships that are strongly supported by primary evidence from those that appear mainly through secondary reporting.

Finally, future work could explore the temporal dimension of the corpus. The current analysis collapses documents across time into a single static network. A dynamic network approach could examine how relationships appear, disappear, or become more legally prominent over time. This would be especially useful for distinguishing early social or aviation relationships from later relationships that become visible through litigation and media coverage.

# 6 Conclusion

This report built an aviation-derived relationship network from the Epstein Files and used it to explore how document provenance changes the interpretation of network structure. Aviation-related relationships involving Jeffrey Epstein were used to identify a manageable actor set, but the final network was reconstructed from all documented relationships among those actors. This allowed the analysis to move beyond a simple travel network and examine the broader actor-to-actor structure present in the corpus.

The main finding is that provenance matters. The full network and the legal subnetwork contain many of the same central actors, but they do not give identical pictures of importance. In the full network, prominence reflects relationships supported across media, communication, legal, and financial sources. In the legal network, prominence is restricted to relationships supported by court filings, transcripts, public records, and other legal-source documents. This shift changes the relative position of several actors. Virginia Roberts Giuffre and Alan M. Dershowitz become more prominent in the legal network, while several political figures become less prominent relative to the full network.

The legal network is smaller, but it is not simply a collapsed version of the full graph. It remains connected, has a comparable diameter, and preserves the broad core of the reconstructed aviation network. This suggests that legal-source documents capture a smaller but still cohesive relational structure. At the same time, the changes in PageRank and betweenness show that removing non-legal relationships alters how actors function within that structure.

The broader conclusion is that extracted relationship networks should not be treated as neutral outputs of a database. Choices about entity consolidation, actor selection, edge aggregation, and provenance filtering shape the final network and therefore shape the interpretation. In this project, retaining document provenance made it possible to compare different evidential views of the same actor set. The legal network should not be treated as a ground-truth version of the graph, but it provides a useful alternative lens through which to examine relationships recorded in the corpus.

Overall, the analysis shows the value of a provenance-aware approach to network construction. Rather than asking only who is connected to whom, the method also asks what kind of documents support those connections. This distinction is especially important when working with heterogeneous and sensitive document collections, where media reporting, legal testimony, public records, and correspondence can produce different versions of the same underlying network.

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